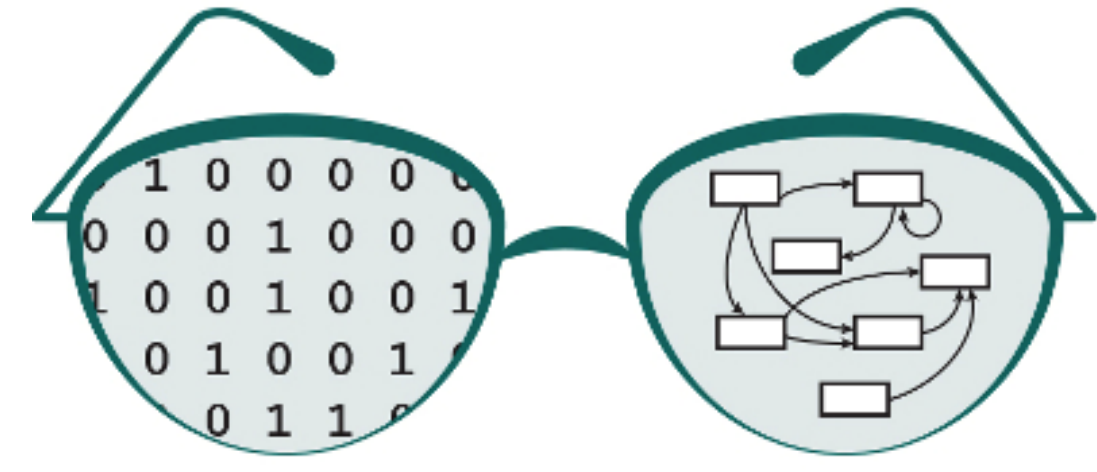


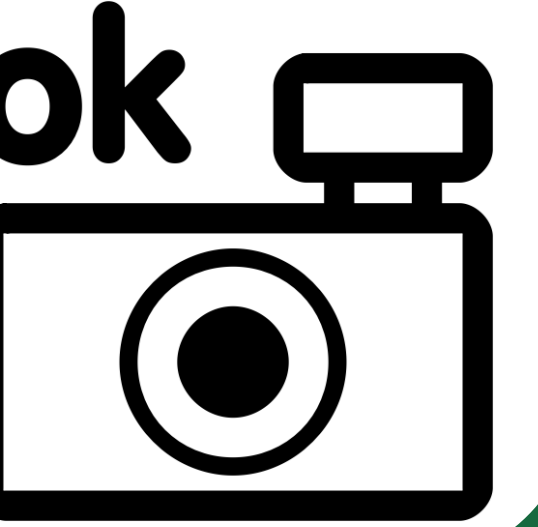
Extending the GRNsight application for visualizing small-to-medium gene regulatory networks to incorporate physical protein-protein interaction data from SGD



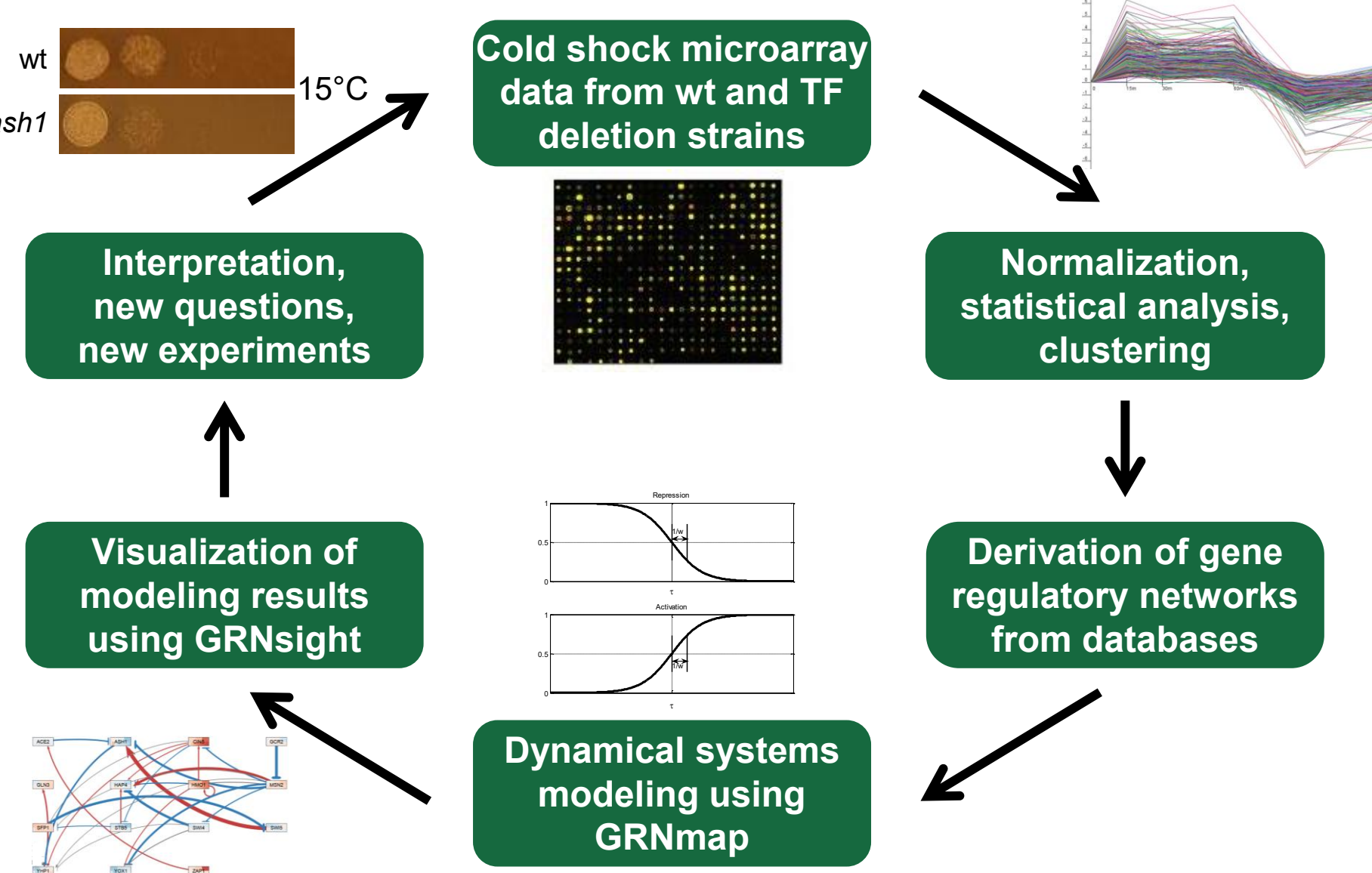
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<http://kdahlquist.github.io/GRNmap/> and <http://dondi.github.io/GRNsight/>

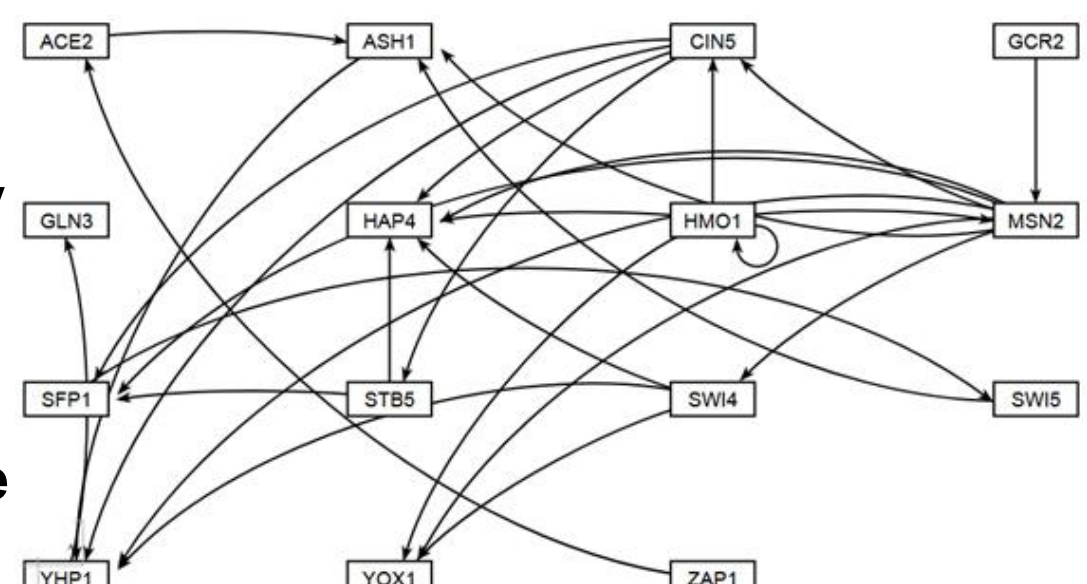


We have previously used a systems biology approach to understand the regulation of the cold shock response in yeast



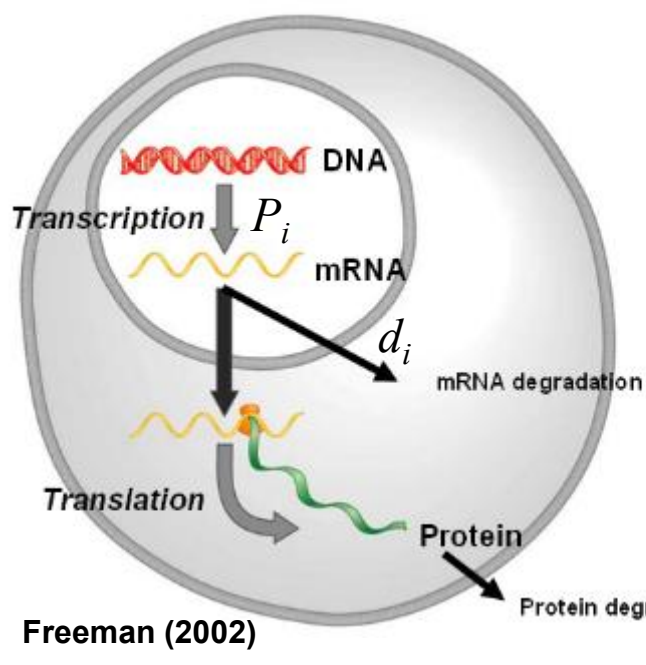
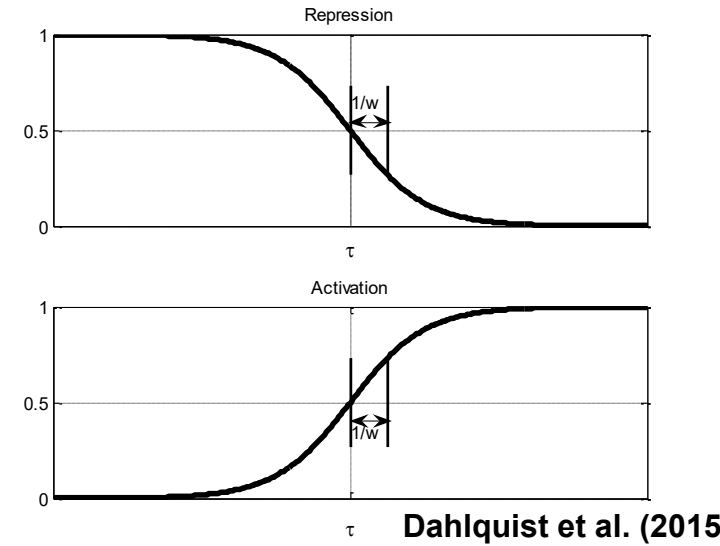
A “medium-scale” gene regulatory network that regulates the cold shock response

- Each node represents one gene encoding a transcription factor.
- Each edge represents a regulatory relationship, either activation or repression, depending on the sign of the weight.
- This network was derived from the YEASTRACT database.



GRNmap: Gene Regulatory Network modeling and parameter estimation

$$\frac{dx_i(t)}{dt} = \frac{P_i}{1 + \exp\left(-\left(\sum_j (w_{ij}x_j(t) - b_i)\right)\right)} - d_i x_i(t)$$



- Model is coded in MATLAB.
- Input Excel workbook contains:
 - Network adjacency matrix
 - Timecourse expression data
 - mRNA degradation rates
- Output workbook contains optimized:
 - Production rates, P
 - Threshold b
 - Weight parameters, w , give the direction (activation or repression) and magnitude of the regulatory relationship

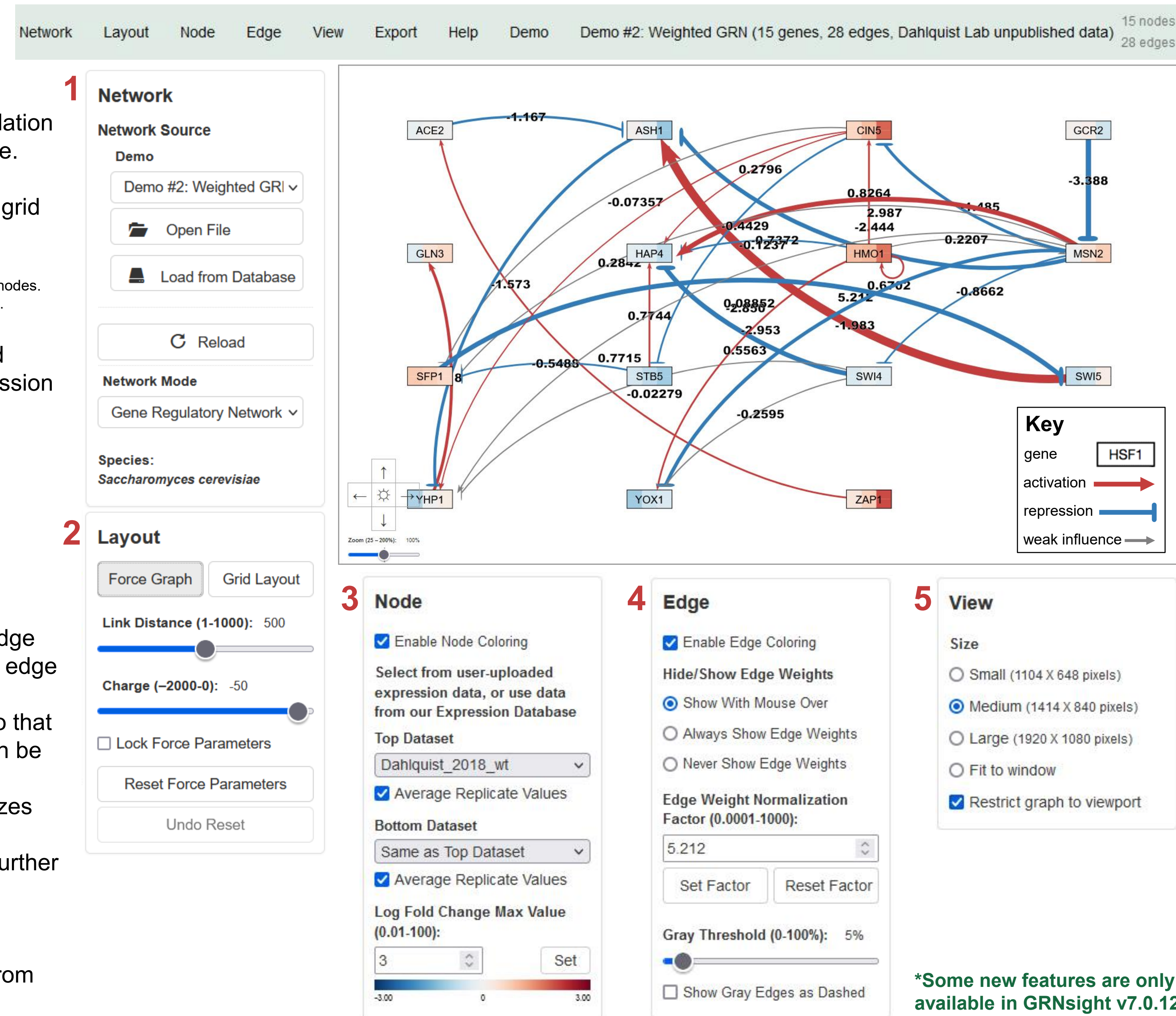
GRNmap produces an Excel workbook with optimized parameters

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
1	col: regulators/shows targets	ABF1	ACE2	AFI1	CIN8	CUP9	FKH1	GTB1	HAL5	HSP1	MAC1	MSN1	MSN4	NRG1	PHO1	RAP1	REB1	ROX1	RPH1	SKN7	YAP1	YAP8
2	ABF1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
3	ACE2	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
4	AFI1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
5	CIN8	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
6	CUP9	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
7	FKH1	0.1950	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
8	GTB1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
9	HAL5	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
10	HSP1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
11	MAC1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
12	MSN1	-2.9707	0	0	0.8380	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
13	MSN4	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
14	NRG1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
15	PHO1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
16	RAP1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
17	REB1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
18	ROX1	0	0	0	-0.9279	0	0	0	0	0	0	0	0	0	0	0	0	0	0.5744	0.4315	-0.5071	0
19	RPH1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
20	SKN7	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
21	YAP1	0	-1.3815	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	-0.4062	0	0
22	YAP8	0	0	0	-0.5912	-0.1293	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0149	-0.3927	0

- Screenshot of the Excel worksheet containing the optimized regulatory weights in the form of an adjacency matrix.
- This format is directly read by GRNsight.

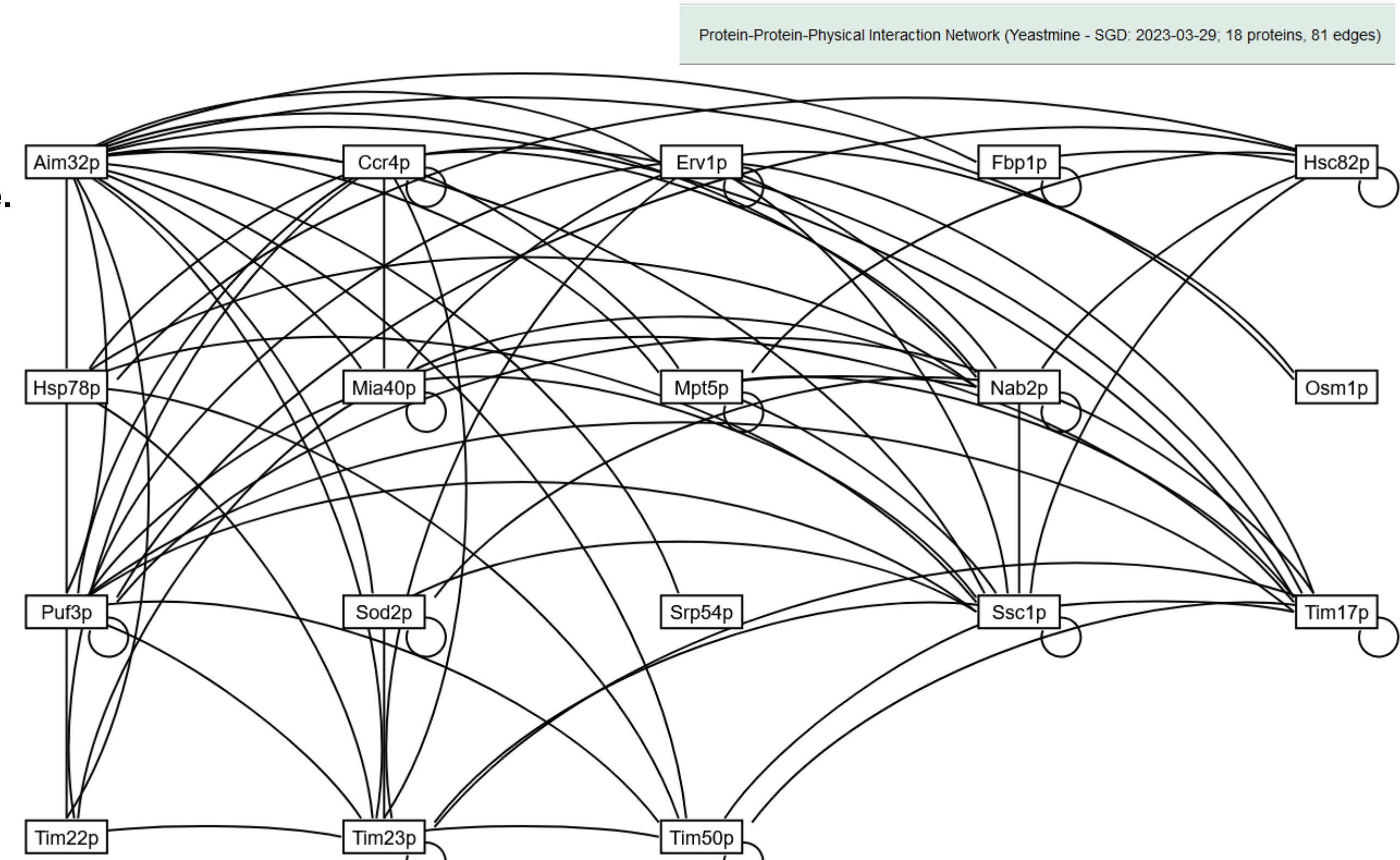
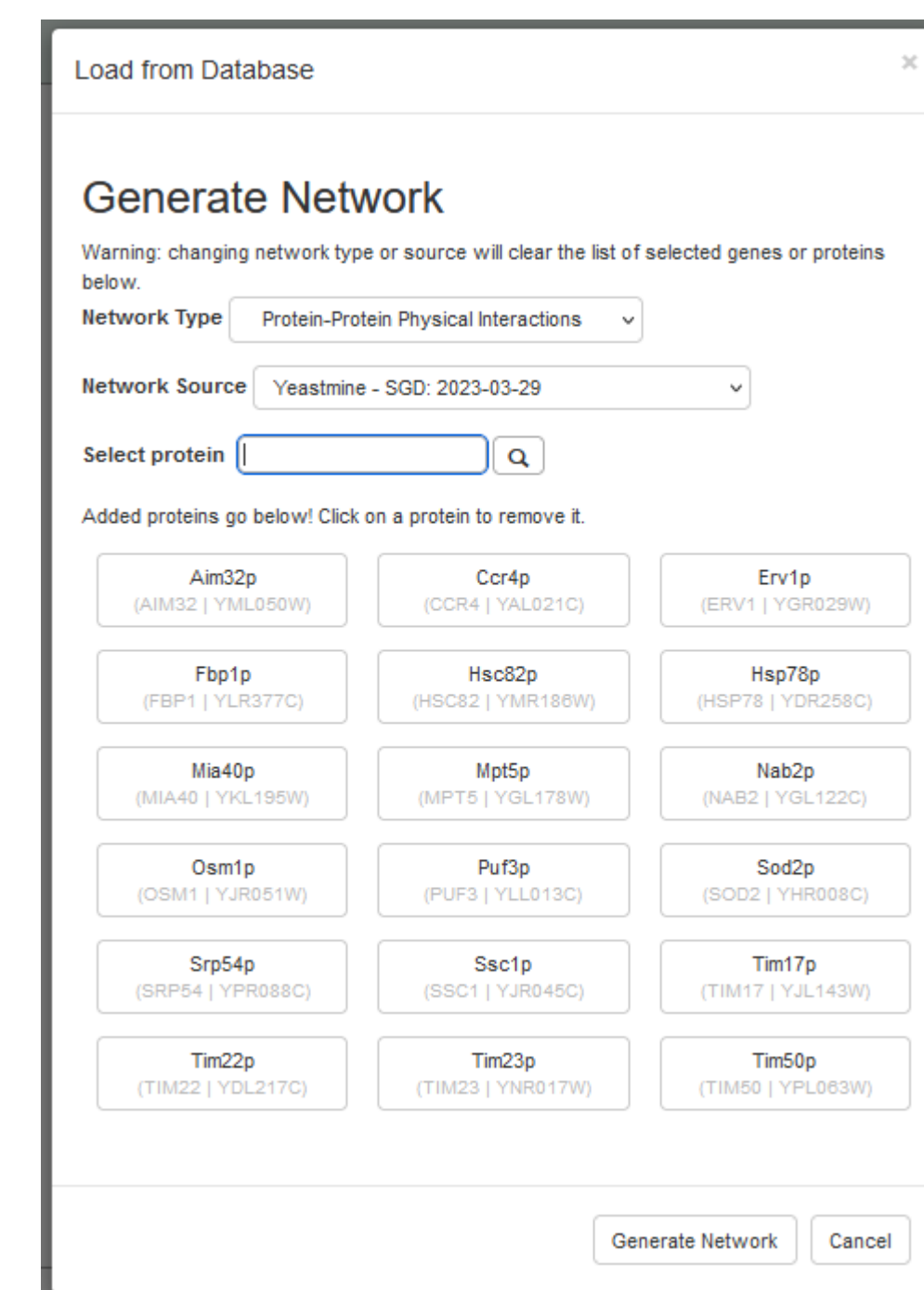
GRNsight, automatically lays out the network graph, coloring the edges based on weight values, and the nodes based on experimental or simulated expression data

- Network**
 - Demo files are provided.
 - Can import and export Excel, SIF, or GraphML files.
 - Can construct a network from SGD regulation or PPI data stored in a backend database.
- Layout**
 - Users can select a force graph layout or grid layout.
 - Force graph parameter sliders:
 - Link distance determines the minimum distance between nodes.
 - Nodes have a charge which repels or attracts other nodes.
- Node**
 - Coloring can be based on user-uploaded expression data or from published expression datasets stored in a backend database.
 - Datasets available:
 - Apweiler et al. 2012, GSE33098
 - Barreto et al. 2012, GSE24712
 - Dahlquist et al. 2018, GSE83656
 - Kitagawa et al. 2002, GSE9336
 - Thorsen et al. 2007, GSE6068
- Edge**
 - Edge coloring can be toggled on and off.
 - Buttons enable the user to always see edge weights, never see edge weights, or see edge weights upon mouseover.
 - Allows user to set normalization factor so that edge thicknesses for different graphs can be rendered on the same scale.
 - Changing the gray threshold deemphasizes weaker edges in the visualization.
 - Presenting gray edges as dashed lines further deemphasizes weak edges.
- View**
 - Multiple viewport sizes available.
 - Graph bounding box can be separated from viewport.
 - Zooming and scrolling available.



GRNsight can now be used to visualize protein-protein physical interaction (PPI) networks

- We previously created a backend database for GRNsight with SGD regulation data gathered from YeastMine.
- We have now included SGD-derived Protein-Protein Physical Interactions (PPI) in the database.
- Users can now select proteins that they want to include in the PPI network from within GRNsight.



- Example of an 18-protein, 81-edge network representing proteins that interact with Aim32p.
- Aim32p is a dual-localized 2Fe-2S mitochondrial protein that functions in redox quality control.
- Unlike GRNs, PPI graphs have undirected edges of uniform thickness and black color and do not display expression data on the nodes.

Goal is to facilitate comparisons between GRNs and PPI graphs for the same phenomenon

- GRN (Yeastmine - SGD: 2023-03-08, 23 genes, 78 edges)
 - A list of transcriptional regulators for the 18 genes of the Aim32p PPI network were obtained from SGD.
 - A total of 64 unique regulators (out of 159 total) were annotated to the 18 Aim32p PPI genes.
 - They were mostly annotations from high-throughput data.
- GRN (Yeastmine - SGD: 2023-03-08, 18 genes, 51 edges)
 - The number of regulators annotated per gene:
 - Minimum: 0 (OSM1)
 - Maximum: 20 (HSC82)
 - Median: 7
 - Average: 9
 - Mode: 4
- GRN (Yeastmine - SGD: 2023-03-08, 11 genes, 24 edges)
 - GRNs were generated from transcriptional regulators annotated in common among the genes.
 - 23 regulators annotated to 3 or more genes
 - 18 regulators annotated to 4 or more genes
 - 11 regulators annotated to 5 or more genes

Future Directions

- Automate the process of toggling between a particular PPI network and the GRN that regulates the genes encoding the proteins in the PPI network, as shown above.
- Update the backend GRNsight GRN and PPI databases with 2024 data from SGD.
- Provide additional time-course expression datasets for users to visualize on the GRN nodes.
 - Students in the Spring 2024 Biology 367 Biological Databases course at LMU will be processing two additional expression datasets for inclusion in GRNsight.

Availability

- GRNsight is free and open to all users at <http://dondi.github.io/GRNsight/>, and there is no login requirement.
- GRNsight code is available under the open source BSD license at our GitHub repository at <https://github.com/dondi/GRNsight>.
- GRNmap MATLAB code and executable can be downloaded from <http://kdahlquist.github.io/GRNmap/downloads.html> under the BSD license.
- Some new GRNsight features are only available in GRNsight v7.0.12 beta.

Acknowledgments

We would like to thank the previous developers on the GRNmap and GRNsight projects, and new students Nikki C. Chun, A'Kaia Phelps, and Cecilia J. Zaragoza.

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